

EEE3027 Integrated Circuit Design

Semester 1 Report

Sahas Talasila *230057896*

CONTENTS

1	Introduction	2
2	NMOS and PMOS I-V Characteristics Lab 1	2
2.1	NMOS Characteristics	2
2.1.1	MOSFET Types	2
2.1.2	Operating Regions of a MOSFET	2
2.2	PMOS characteristics	5
2.3	Beta Calculation	7
2.3.1	Definition	8
2.3.2	Beta Expression	8
2.3.3	Related Terms	8
2.3.4	Importance	8
2.3.5	Question 2A Model Parameter Definitions	9
2.3.6	Question 2B Model Assumptions	9
2.4	How SPICE calculates non linear equations	10
2.4.1	Actual Example of Newton Raphson Method with a Circuit	10
3	Lab 1 part 2: CMOS Inverter and Experiments	12
3.1	CMOS Inverter Diagram and Theory	12
3.2	CMOS Missing SPICE	12
3.3	Transfer Curves	12
3.4	NMOS Pull Up	13
3.5	CMOS Inverter Design	13

Abstract

This is a placeholder for the abstract of the report.

INTRODUCTION

NMOS AND PMOS I-V CHARACTERISTICS LAB 1

2.1 NMOS Characteristics

A transistor is a semiconductor device used to amplify or switch electronic signals and electrical power. It is composed of semiconductor material, usually with at least three terminals for connection to an external circuit. A voltage or current applied to one pair of the transistor's terminals controls the current through another pair of terminals.

Transistor Structure:

- **Source (S):** The terminal through which carriers enter the channel.
- **Drain (D):** The terminal through which carriers leave the channel.
- **Gate (G):** The terminal that modulates the conductivity of the channel.
- **Body (B):** The substrate on which the transistor is built, often connected to the source.
- **Channel:** The region between the source and drain where current flows when the transistor is on.
- **Oxide Layer:** An insulating layer between the gate and the channel, typically made of silicon dioxide (SiO_2).



Figure 1: Basic structure of an NMOS and PMOS transistor.

2.1.1 MOSFET Types

- **N-channel (NMOS):** electrons are charge carriers.
- **P-channel (PMOS):** holes are charge carriers.

2.1.2 Operating Regions of a MOSFET

Region	Condition (N-channel)	Description
Cutoff	$V_{GS} < V_{th}$	OFF (no current)
Triode (Linear)	$V_{GS} > V_{th}, V_{DS} < V_{GS} - V_{th}$	Acts as variable resistor
Saturation (Active)	$V_{GS} > V_{th}, V_{DS} \geq V_{GS} - V_{th}$	Current source (amplifier region)

A MOS transistor operates in two main regions depending on the drain-source voltage (V_{DS}) and the gate-source voltage (V_{GS}). In *Figure 2*, we can see the NMOS Sweep for a few specified V_{GS} values.

It is important to understand the difference between electrons and holes in semiconductor physics and for transistor design:

- **Electrons**

- Negative charge: $-e$
- Real particles in the **conduction band**
- Move toward the positive terminal

- **Holes**

- Positive charge: $+e$
- "Missing electrons" in the **valence band**
- Act like positive particles moving toward the negative terminal

Explain the code used to generate the plots,

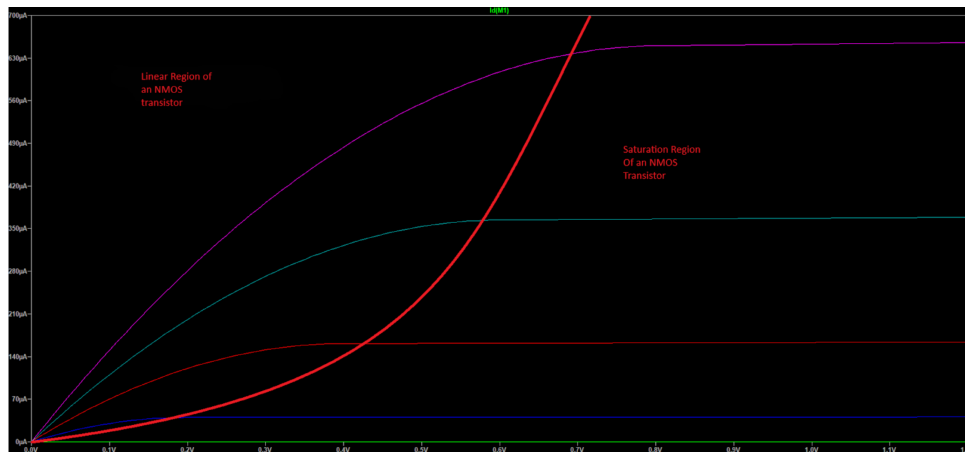


Figure 2: NMOS I-V Characteristics showing Linear and Saturation regions for different V_{GS} values.

Linear Region

This occurs when:

$$V_{DS} < V_{GS} - V_T \quad (1)$$

In this region, the transistor behaves like a voltage-controlled resistor. The drain current is given by *Equation 11*:

$$I_D = \beta \left[(V_{GS} - V_T)V_{DS} - \frac{1}{2}V_{DS}^2 \right] \quad (2)$$

where, if we refer to *Equation 10* again:

$$\beta = \frac{W}{L} \mu_n C_{ox} = \frac{W}{L} K_P$$

Initially, the current I_D increases approximately linearly with V_{DS} .

Saturation Region

This occurs when:

$$V_{DS} \geq V_{GS} - V_T \quad (3)$$

In this region, the channel is *pinched off* near the drain, and the drain current becomes almost independent of V_{DS} :

$$I_D = \frac{\beta}{2} (V_{GS} - V_T)^2 (1 + \lambda V_{DS}) \quad (4)$$

The calculations for each curve are shown below:

- **EXPECTED CURVES:**

- **Curve 1 ($V_{GS} = 0.4 \text{ V}$):**

- * Barely turns on ($V_{GS} = V_{th}$)
- * Very low current
- * Saturation at $I_{DS} \approx 0 \text{ mA}$

- **Curve 2 ($V_{GS} = 0.6 \text{ V}$):**

- * $V_{DSsat} = 0.6 - 0.4 = 0.2 \text{ V}$
- * Saturation current (approx) = 0.04 mA
- * $I_{DS,sat} = \frac{\beta}{2} (V_{GS} - V_{th})^2 = 1 \text{ m} \times (0.2)^2 = 0.04 \text{ mA}$

- **Curve 3 ($V_{GS} = 0.8 \text{ V}$):**

- * $V_{DSsat} = 0.8 - 0.4 = 0.4 \text{ V}$
- * Saturation current (approx) = 0.16 mA
- * $I_{DS,sat} = 1 \text{ m} \times (0.4)^2 = 0.16 \text{ mA}$

- **Curve 4 ($V_{GS} = 1.0 \text{ V}$):**

- * $V_{DSsat} = 1.0 - 0.4 = 0.6 \text{ V}$
- * Saturation current (approx) = 0.36 mA
- * $I_{DS,sat} = 1 \text{ m} \times (0.6)^2 = 0.36 \text{ mA}$

- **Curve 5 ($V_{GS} = 1.2 \text{ V}$):**

- * $V_{DSsat} = 1.2 - 0.4 = 0.8 \text{ V}$

- * Saturation current (approx) = 0.64 mA
- * $I_{DS,sat} = 1 \text{ m} \times (0.8)^2 = 0.64 \text{ mA}$
- **Each curve:**
 - * Linear region: 0 to $(V_{GS} - V_{th})$
 - * Saturation region: $(V_{GS} - V_{th})$ to 1.2 V
 - * Higher $V_{GS} \Rightarrow$ higher saturation current
 - * Saturation current $\propto (V_{GS} - V_{th})^2$

2.2 PMOS characteristics

As we have already talked about the theory behind NMOS transistors, we will now look at PMOS transistors, this section will be shorter as the underlying theory is the same.

- **Difference Between NMOS and PMOS:**
 - **Type of Charge Carrier:**
 - * NMOS: Uses electrons as majority carriers.
 - * PMOS: Uses holes as majority carriers.
 - **Polarity of Operation:**
 - * NMOS: Turns on when $V_{GS} > V_{th}$ (positive gate voltage).
 - * PMOS: Turns on when $V_{GS} < V_{th}$ (negative gate voltage).
 - **Channel Type:**
 - * NMOS: n-type channel.
 - * PMOS: p-type channel.
 - **Symbol:**
 - * NMOS: Arrow points *out* of the source.
 - * PMOS: Arrow points *into* the source.
 - **Carrier Mobility:**
 - * NMOS: Higher mobility (electrons move faster).
 - * PMOS: Lower mobility (holes move slower).
 - **Threshold Voltage (V_{th}):**
 - * NMOS: Positive threshold voltage.
 - * PMOS: Negative threshold voltage.
 - **Logic Level Representation:**
 - * NMOS: Logic '1' (high voltage) turns it ON.
 - * PMOS: Logic '0' (low voltage) turns it ON.

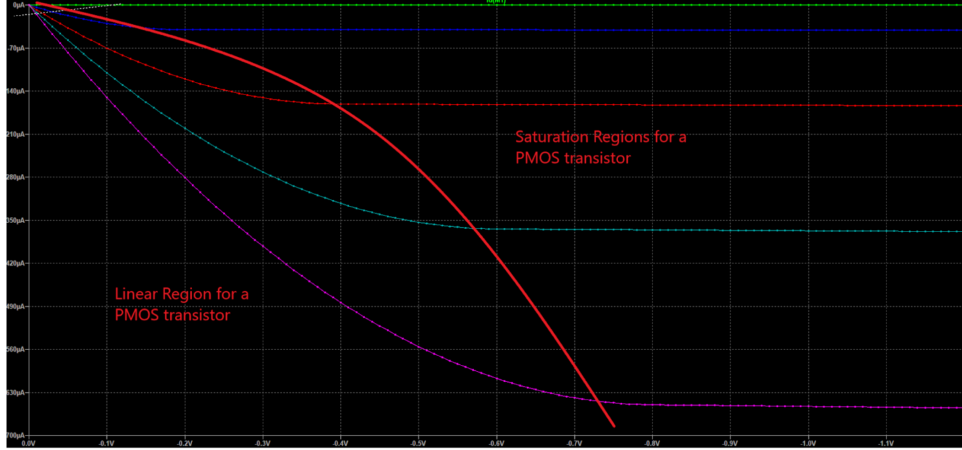


Figure 3: PMOS I-V Characteristics showing Linear and Saturation regions for different V_{SG} values.

Explain the code to generate the plots (just a small difference),

Explain the plots. (plot of I_D vs V_{DS})

For a PMOS transistor, all voltages are **negative** relative to the NMOS device. The key differences are:

- V_{SG} (source-to-gate voltage) replaces V_{GS}
- V_{SD} (source-to-drain voltage) replaces V_{DS}
- The drain current I_D flows from the **source to the drain**, opposite to that of the NMOS

Linear Region

This occurs when:

$$V_{SD} < V_{SG} - |V_T|$$

The transistor behaves like a voltage-controlled resistor. The drain current is given by:

$$I_D = \beta \left[(V_{SG} - |V_T|)V_{SD} - \frac{1}{2}V_{SD}^2 \right]$$

where:

$$\beta = \frac{W}{L} \mu_p C_{ox} = \frac{W}{L} K_P$$

Initially, I_D increases approximately linearly with V_{SD} .

Saturation Region

This occurs when:

$$V_{SD} \geq V_{SG} - |V_T|$$

In this region, the channel is *pinched off* near the drain, and the current becomes almost independent of V_{SD} :

$$I_D = \frac{\beta}{2} (V_{SG} - |V_T|)^2 (1 + \lambda V_{SD})$$

The small increase in I_D with V_{SD} is due to **channel-length modulation**, represented by λ .

Finally, show the calculations for each curve. Example shown below

- **EXPECTED CURVES:**

- **Curve 1 ($V_{GS} = 0.4 \text{ V}$):**

- * Barely turns on ($V_{GS} = V_{th}$)
- * Very low current
- * Saturation at $I_{DS} \approx 0 \text{ mA}$

- **Curve 2 ($V_{GS} = 0.6 \text{ V}$):**

- * $V_{DSsat} = 0.6 - 0.4 = 0.2 \text{ V}$
- * Saturation current (approx) = 0.04 mA
- * $I_{DS,sat} = \frac{\beta}{2}(V_{GS} - V_{th})^2 = 1 \text{ m} \times (0.2)^2 = 0.04 \text{ mA}$

- **Curve 3 ($V_{GS} = 0.8 \text{ V}$):**

- * $V_{DSsat} = 0.8 - 0.4 = 0.4 \text{ V}$
- * Saturation current (approx) = 0.16 mA
- * $I_{DS,sat} = 1 \text{ m} \times (0.4)^2 = 0.16 \text{ mA}$

- **Curve 4 ($V_{GS} = 1.0 \text{ V}$):**

- * $V_{DSsat} = 1.0 - 0.4 = 0.6 \text{ V}$
- * Saturation current (approx) = 0.36 mA
- * $I_{DS,sat} = 1 \text{ m} \times (0.6)^2 = 0.36 \text{ mA}$

- **Curve 5 ($V_{GS} = 1.2 \text{ V}$):**

- * $V_{DSsat} = 1.2 - 0.4 = 0.8 \text{ V}$
- * Saturation current (approx) = 0.64 mA
- * $I_{DS,sat} = 1 \text{ m} \times (0.8)^2 = 0.64 \text{ mA}$

- **Each curve:**

- * Linear region: 0 to $(V_{GS} - V_{th})$
- * Saturation region: $(V_{GS} - V_{th})$ to 1.2 V
- * Higher $V_{GS} \Rightarrow$ higher saturation current
- * Saturation current $\propto (V_{GS} - V_{th})^2$

2.3 Beta Calculation

Explain what beta is, and how to calculate it.

Beta (β): A constant that represents the gain factor of a MOSFET, given by *Equation 9*. It determines how much current the transistor can conduct for a given gate voltage.

2.3.1 Definition

$$I_D = \frac{1}{2}\beta(V_{GS} - V_{th})^2 \quad (\text{saturation region}) \quad (5)$$

$$I_D = \beta \left[(V_{GS} - V_{th})V_{DS} - \frac{V_{DS}^2}{2} \right] \quad (\text{linear region}) \quad (6)$$

2.3.2 Beta Expression

Beta can also be expressed by *Equation 9*:

$$\boxed{\beta = \mu_n C_{ox} \frac{W}{L}} \quad (7)$$

where:

- μ_n = carrier mobility
- $C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$ = oxide capacitance per unit area
- W = gate width
- L = gate length

2.3.3 Related Terms

$$k' = \mu_n C_{ox}, \quad \beta = k' \frac{W}{L}$$

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \beta(V_{GS} - V_{th})$$

2.3.4 Importance

- Determines how strongly a MOSFET amplifies a signal.
- Affects gain, speed, and current drive.
- Critical for matching and biasing in analogue ICs.

Explain the code again and then show a plot of beta and then gradient calculation.

Explain what the gradient means.

Show image with gradients, then compare theoretical and experimental values.

Explain the relationship between μ , C_{ox} , W and L , and how changing W and L will affect beta.

Show plot with l times 2 and how gradient has halved, referring back to the beta equation and the equivalent versions.

Show how VGS also affects beta. (has to be in saturation region for this to be true), otherwise it is not valid. Show plot with VGS changed and how beta changes accordingly. Refer back to beta equation again.

Beta vs width or length table

2.3.5 Question 2A Model Parameter Definitions

VTO (Threshold Voltage): Symbol: V_T or V_{th} . Units: Volts (V).

Zero-bias threshold voltage - the minimum gate-source voltage required to form a conducting inversion channel. Below this voltage, the transistor is in cutoff. Determined by gate material work function, oxide thickness, substrate doping, and interface charges. In our simulation: VTO = 0.4V (NMOS), -0.4V (PMOS).

KP (Transconductance Parameter): Symbol: k' or μC_{ox} . Units: A/V².

Intrinsic transconductance parameter representing the product of carrier mobility (μ) and gate oxide capacitance per unit area (C_{ox}):

$$KP = \mu \cdot C_{ox} = \mu \cdot \frac{\epsilon_{ox}}{t_{ox}} \quad (8)$$

Determines current drive capability. The gain factor is $\beta = (W/L) \cdot KP$, so $I_D \propto KP$. NMOS typically has 2-3 $\times$ higher KP than PMOS due to higher electron mobility. In our simulation: KP = 200 μ A/V² for both types.

LAMBDA (Channel-Length Modulation): Symbol: λ . Units: V⁻¹. Default: 0.0.

Channel-length modulation coefficient accounting for effective channel shortening as V_{DS} increases in saturation. Without LAMBDA, saturation curves would be perfectly flat (infinite output resistance). With LAMBDA:

$$I_D(sat) = \frac{\beta}{2} (V_{GS} - V_T)^2 (1 + \lambda V_{DS}) \quad (9)$$

Output resistance: $r_o = 1/(\lambda I_D)$. Inversely proportional to channel length: $\lambda \propto 1/L$. Analogous to inverse Early voltage in BJTs. In our simulation: $\lambda = 0.02$ V⁻¹, causing $\sim 2.4\%$ current increase across 1.2V V_{DS} range.

2.3.6 Question 2B Model Assumptions

The standard textbook equation for saturation current is:

$$I_D = \frac{\beta}{2} (V_{GS} - V_T)^2 \quad \text{where } \beta = \frac{W}{L} \cdot \mu_n C_{ox} \quad (10)$$

The complete SPICE Level 1 equation is:

$$I_D = \frac{KP}{2} \cdot \frac{W_{eff}}{L_{eff}} \cdot (V_{GS} - VTO)^2 \cdot (1 + LAMBDA \cdot V_{DS}) \quad (11)$$

To simplify the SPICE model to match textbook equations, the following assumptions are made:

1. LAMBDA = 0 (or very small): The term $(1 + \lambda V_{DS})$ is set to 1, neglecting channel-length modulation. This assumes perfectly flat I-V curves in saturation with infinite output resistance. In our simulation, $\lambda = 0.02$ causes only a $\sim 2.4\%$ variation, which is small but non-zero.

2. No body effect ($\gamma = 0$ or $V_{SB} = 0$): The body effect modifies threshold voltage as:

$$V_T = V_{T0} + \gamma(\sqrt{|2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|}) \quad (12)$$

When bulk and source are both at ground ($V_{SB} = 0$), this term vanishes and $V_T = V_{T0}$, which is our case.

3. Long-channel approximation: Velocity saturation is neglected, assuming drift velocity remains proportional to electric field. Valid for $L \geq 1\mu\text{m}$. Short-channel effects (DIBL, threshold roll-off) are ignored.

4. No mobility degradation: Carrier mobility μ is assumed constant, independent of V_{GS} . In reality, high vertical electric fields reduce mobility, but Level 1 ignores this.

5. Ideal effective dimensions: $W_{eff} = W$ and $L_{eff} = L$ (no parasitic effects). When $LD = WD = XL = XW = DEL = 0$, drawn and effective dimensions are equal.

6. Subthreshold conduction ignored: For $V_{GS} < V_T$, Level 1 assumes $I_D = 0$. In reality, exponential subthreshold current exists: $I_D \propto \exp(V_{GS}/nV_t)$.

Verification of equivalence: If $KP = \mu_n C_{ox}$, then:

$$\beta_{textbook} = \frac{W}{L} \cdot \mu_n C_{ox} = \frac{W}{L} \cdot KP = \beta_{SPICE} \quad (13)$$

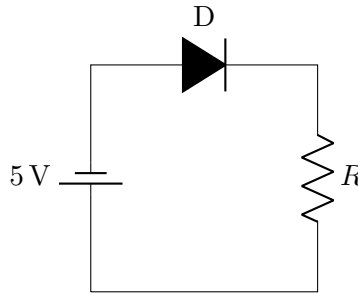
With $LAMBDA = 0$, $V_{SB} = 0$, and ideal geometry, the equations become identical.

2.4 How SPICE calculates non linear equations

Include brief theory on non linear equations in circuits.

2.4.1 Actual Example of Newton Raphson Method with a Circuit

Consider the series circuit: a voltage source V_s , a resistor R and a diode D in series. We wish to find the diode voltage V_d at DC (steady state).



Kirchhoff's Voltage Law (KVL) around the loop gives

$$V_s - IR - V_d = 0 \quad (14)$$

and the diode follows the Shockley equation

$$I = I_s \left(e^{\frac{V_d}{nV_t}} - 1 \right) \quad (15)$$

Substituting the diode law into KVL yields the nonlinear scalar equation for V_d :

$$f(V_d) \equiv V_s - R I_s \left(e^{\frac{V_d}{nV_t}} - 1 \right) - V_d = 0. \quad (16)$$

To solve $f(V_d) = 0$ with Newton–Raphson, form the iteration

$$V_d^{(k+1)} = V_d^{(k)} - \frac{f(V_d^{(k)})}{f'(V_d^{(k)})}. \quad (17)$$

We need the derivative $f'(V_d)$. Differentiating (3) gives

$$f'(V_d) = -R I_s \frac{1}{nV_t} e^{\frac{V_d}{nV_t}} - 1. \quad (18)$$

Thus each Newton step can be written explicitly as

$$V_d^{(k+1)} = V_d^{(k)} - \frac{V_s - R I_s \left(e^{\frac{V_d^{(k)}}{nV_t}} - 1 \right) - V_d^{(k)}}{-R I_s \frac{1}{nV_t} e^{\frac{V_d^{(k)}}{nV_t}} - 1}. \quad (19)$$

A practical step-by-step procedure:

1. Choose device and circuit parameters V_s, R, I_s, n, V_t and an initial guess $V_d^{(0)}$ (a typical choice is 0.6–0.8 V for silicon diodes, or $V_s/2$ if unsure).
2. Evaluate $f(V_d^{(k)})$ using (3) and $f'(V_d^{(k)})$ using (4).
3. Compute the Newton update by (5) to obtain $V_d^{(k+1)}$.
4. Check convergence: accept $V_d^{(k+1)}$ if

$$|V_d^{(k+1)} - V_d^{(k)}| < \varepsilon \quad \text{or} \quad |f(V_d^{(k+1)})| < \delta, \quad (20)$$

where ε, δ are tolerances (e.g. 10^{-6} V or a small current error).

5. If not converged, set $k \leftarrow k+1$ and repeat from step 2. Include a maximum iteration limit to avoid infinite loops.

Compare bisection method to newton raphson method and explain why newton raphson is better for SPICE.

Add flowchart//pseudocode of how SPICE solves non linear equations using newton raphson method.

3.1 CMOS Inverter Diagram and Theory

CMOS Inverter: A CMOS inverter consists of a PMOS and an NMOS transistor connected in series.

- The **PMOS** is connected to the supply voltage (V_{DD}).
- The **NMOS** is connected to ground.
- Both gates are connected together to form the input (V_{in}).
- The drains are connected together to form the output (V_{out}).

Operation:

- When V_{in} is low (0): PMOS is ON, NMOS is OFF $\Rightarrow V_{out} = V_{DD}$ (logic 1).
- When V_{in} is high (V_{DD}): PMOS is OFF, NMOS is ON $\Rightarrow V_{out} = 0$ (logic 0).

Thus, the CMOS inverter **inverts the input logic level**.

png of CMOS inverter circuit. (need to send from PC)

3.2 CMOS Missing SPICE

Explain the missing SPICE code and how it works.

Explain the additions and annotations made to the original code.

3.3 Transfer Curves

Explain what the transfer curves and how they were generated.

Point out the plots and explain the different regions of operation for PMOS and NMOS.

Explain when NMOS is on and when PMOS is on/triode/saturation.

Analysis table showing what happens at each voltage level.

Point	V_{in} (V)	V_{out} (V)	V_{GSN} (V)	V_{DSN} (V)	NMOS Mode	V_{GSP} (V)	V_{DSP} (V)
A	0	≈ 5.0	0	≈ 5.0	OFF	-5.0	≈ 0
B	1	≈ 4.0	1.0	≈ 4.0	SAT (just on)	-4.0	≈ -1.0
C	2.5	≈ 2.5	2.5	≈ 2.5	SAT	-2.5	≈ -2.5
D	4	≈ 1.0	4.0	≈ 1.0	TRIODE (LINEAR)	-1.0	≈ -4.0
E	5	≈ 0	5.0	≈ 0	TRIODE	0	-5.0

Table 1: CMOS Inverter Operation Points

CMOS Inverter Transfer Curves: The transfer curve of a CMOS inverter represents how the output voltage (V_{out}) changes in response to the input voltage (V_{in}). It is obtained by sweeping V_{in} from 0 to V_{DD} and measuring the corresponding V_{out} for each input value. The curve shows the inverter's switching behavior — how it transitions from logic high to logic low.

Operation Regions:

- **Region 1:** $V_{in} = 0$ V (**Logic 0 Input**)
 - NMOS: $V_{GSN} = 0 \Rightarrow$ **OFF**
 - PMOS: $V_{GSP} = -V_{DD} \Rightarrow$ **ON (Triode region)**
 - Output: $V_{out} \approx V_{DD}$ (Logic 1)
- **Region 2:** V_{in} slightly increases (0 V $< V_{in} < V_{th,n}$)
 - NMOS: Still **OFF**
 - PMOS: **ON** and in **saturation** or **triode** region
 - Output: Remains near V_{DD}
- **Region 3:** $V_{in} \approx V_{th,n}$ to $V_{DD} - |V_{th,p}|$ (**Transition region**)
 - Both transistors conduct simultaneously.
 - NMOS: **Saturation** region.
 - PMOS: **Saturation** region.
 - Output: Rapidly falls from V_{DD} toward 0 V.
 - This is the **switching point** of the inverter, where $V_{in} \approx V_{out}$.
- **Region 4:** V_{in} near V_{DD} (**Logic 1 Input**)
 - NMOS: $V_{GSN} = V_{DD} \Rightarrow$ **ON (Triode region)**
 - PMOS: $V_{GSP} = 0 \Rightarrow$ **OFF**
 - Output: $V_{out} \approx 0$ V (Logic 0)

3.4 NMOS Pull Up

Explain the modifications made to the CMOS inverter to create an NMOS pull up.

Show the SPICE code used.

Explain the transfer curves generated and compare to the CMOS inverter.

Show the differences and why they occur.

Show the calculations for the threshold voltages of both NMOS and PMOS in the circuit, and explain how they were calculated and how they affect the operation of the inverter.

Show how the NMOS pull up is worse than the CMOS inverter.

Answer Questions from lab brief.

3.5 CMOS Inverter Design

Explain the requirements for the CMOS inverter design and what they mean.